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Assignment 5: Unit Commitment

# Meeting	Location	Content
1st Session	N3815	Introduction of assignment; start with training stage 1
Time until 2nd Meeting	Home	Work on training stage 1
2nd Session	N3815	Work on competition stage 2; competition; presentations
Until next assignment	Home	Work on report

Background

This exercise is about coordinating electrical generators in the most efficient way, while considering some constraints. More precisely, for a given set of dispatchable and non-dispatchable generators[1], we aim to define a generation schedule for a specific time period, such that the generation costs are as low as possible. In addition, we must ensure that we meet the electricity demand and consider certain generation constraints. Such a task is typically formulated as a mixed-integer nonlinear programming problem.

For this exercise, we will use the software package *pyomo*, which is a Python-based open-source package for modeling structured optimization problems. To run the optimization, we will use the NEOS server as in Assignment 4.

The aim of the assignment is to evaluate

- your understanding of how power system operators minimize operational cost,
- your ability to rigorously formulate optimization problems,
- your ability to apply the pyomo framework to this task.

The expected outcome of the assignment includes

- a report of maximum 2 pages (excluding appendices),
- your optimization results as .json file.

The following aspects of your work will be graded:

- Formulation of optimization problem (correctness)
- Ability to work with pyomo
- Evaluation and interpretation of optimal solution
- Discussion (what went well, what went wrong, how your solution could be improved - in your opinion, etc.)
- Competition ranking

[1] Dispatchable generators are power plants that can be scheduled to generate electricity on demand, while non-dispatchable generators produce electricity intermittently and their output cannot be controlled, such as solar or wind power.

- General presentation of the assignment report (flow of information, quality of English, quality of illustrations, etc.)

Stage 1 – Training

We will now approach our first unit commitment problem to develop the code for the final competition task. We aim to maximize the economic efficiency (i.e., minimize the generation costs) of a fleet of generators (both dispatchable and non-dispatchable) over the course of 24 hours. We have $n = 3$ dispatchable generators, and one wind and one solar power plant. The non-dispatchable part of the power generation is provided as forecasts with associated probabilities and its generation is free of costs. The cost function of dispatchable generator $i \in \{1,2,3\}$ for a generation of p_i is given as

$$C(p_i) = a_{2,i}{p_i}^2 + a_{1,i}p_i.$$

Once a dispatchable generator $i \in \{1,2,3\}$ is switched on, it must produce at least p_{min} and at most p_{max}.

You can download the following three files on Moodle:

- **DataB_generatorparams.csv**: Artificial cost parameter data $a_1, a_2 \in R^n$ and the minimal and maximal generation per dispatchable generator, $p_{min}, p_{max} \in R^n$,
- **DataB_hourlyload.csv**: Load per hour $t = 0, \dots, 23$,
- **DataB_nondispgeneration.csv**: Discrete probability distribution per hour of the non-dispatchable generators wind and solar. We assume that the generation between the hours is independent.

Your task is to find for each hour $t = 0, \dots, 23$,

- the on/off schedule for each dispatchable generator $u_{i,t} \in \{0,1\}$,
- the on/off schedule for the non-dispatchable generators $u_{wind,t}, u_{solar,t} \in \{0,1\}$ and
- the generation of each dispatchable generator $p_{i,t} \in R_{\geq 0}$,

such that

- overall costs are minimized,
- the minimal/maximal generation constraint is fulfilled,
- the overall generation is as close to the actual load as possible.

Violations of constraints will be penalized with additional costs. The smaller the overall costs, the better is your solution.

Note that due to uncertainty in the non-dispatchable generators, formulating the optimization problem is neither straightforward, nor unique.

Please state your result as a .json file. The file must have the following form:

```
{       'name': your team's name as string,
        'task': letter of task, in this case 'B' as string,
```

`'u1'`: binary list of length 24 with on/off schedule for dispatchable generator 1,

`'u2'`: binary list of length 24 with on/off schedule for dispatchable generator 2,

`'u3'`: binary list of length 24 with on/off schedule for dispatchable generator 3,

`'p1'`: list of length 24 with generation schedule for dispatchable generator 1,

`'p2'`: list of length 24 with generation schedule for dispatchable generator 2,

`'p3'`: list of length 24 with generation schedule for dispatchable generator 3,

`'uwind'`: binary list of length 24 with on/off schedule for wind power plant,

`'usolar'`: binary list of length 24 with on/off schedule for solar power plant`}`

Make sure that all list entries of `p` and `u` are numeric! Note that the performance of your solution depends on the realization of the random non-dispatchable generator production. For the competition in Stage 2, your solution will be evaluated using 10000 scenarios, which were randomly drawn according to the given probability distributions and considering the average performance. Each solution will be evaluated with the same 10000 scenarios. As this should come very close to a fair evaluation, but is not perfectly fair, we will give the same score to teams whose results are very close to each other.

You can voluntarily submit your stage 1 solution to Moodle. In the beginning of session 2, the tutors will look at this voluntary submission to let you know whether you submitted in the right format. This option serves solely the purpose of making sure that your submission format is correct, not to help you get a feeling whether you have a good solution approach or not. It is part of your task to find a good way to evaluate your solution yourselves.

Stage 2 – Competition

We will now apply the code we developed in Task B to new data. This time, we are working with real hourly load data from the 7th of March 2023 in Germany (data source: smard.de). The generator data is again artificial, but differs from Task B. The cost function and constraints remain the same.

Look at the data and modify your code, such that it works on the new data set.

Please upload your result in form of .json file to moodle. The file must have the following form:

`{` `'name'`: your team's name as string,

`'task'`: letter of task, in this case 'C' as string,

`'u1'`: binary list of length 24 with on/off schedule for dispatchable generator 1,

`'u2'`: binary list of length 24 with on/off schedule for dispatchable generator 2,

`'u3'`: binary list of length 24 with on/off schedule for dispatchable generator 3,

`'u4'`: binary list of length 24 with on/off schedule for dispatchable generator 4,

`'p1'`: list of length 24 with generation schedule for dispatchable generator 1,

`'p2'`: list of length 24 with generation schedule for dispatchable generator 2,

`'p3'`: list of length 24 with generation schedule for dispatchable generator 3,

`'p4'`: list of length 24 with generation schedule for dispatchable generator 4,

`'uwind'`: binary list of length 24 with on/off schedule for wind power plant,

`'usolar'`: binary list of length 24 with on/off schedule for solar power plant`}`

Penalties

We will assign the following penalties for constraint violations:

- Non-binary `u`: 10000€ per non-binary value
- Min/Max constraints: [your absolute deviation from constraint] * 1000€
- Load mismatch: [your absolute deviation from load profile] * 1000€

Naming of Submissions

Each submission must be named according to the following pattern:

- Competition: f"Competition_5_Group_{Group_letter}.json“
- Presentation: f"Presentation_5_Group_{Group_letter}.pdf“
- Report and Code: f"Report_Code_5_Group_{Group_letter}.zip“

As an example for the group submissions of Group E:

- “Competition_5_Group_E.json”
- “Presentation_5_Group_E.pdf”
- "Report_Code_5_Group_E.zip“

Appendix: Short description of data and various files

The files to be used for this assignment provide all necessary generator information and the hourly load.

DataB_generatorparams.csv

This file contains the artificial cost and constraint parameters of the dispatchable generators.

Variables (columns):

- generator: number of generator (1-3)
- a1: linear cost parameter a_1
- a2: quadratic cost parameter a_2
- pmin: minimal generation p_{min} [kWh]
- pmax: maximal generation p_{max} [kWh]

DataB_hourlyload.csv

This file contains the artificial electricity demand per hour for one day.

Variables (columns):

- hour: hour 0-23
- load: artificial load [kWh]

DataB_nondispgeneration.csv

This file contains an artificial discrete probability distribution of the non-dispatchable generators.

Variables (columns):

- hour: block wise hour column, contains the values 0-5, 6-10, 11-14, 15-19, 20-23
- generation wind: four distinct possible wind generations for each hour block [kWh]
- probability wind: probabilities corresponding to 'generation wind'
- generation solar: four solar generations for each hour block [kWh]
- probability solar: probabilities corresponding to 'generation solar'

DataC_generatorparams.csv

This file contains the artificial cost and constraint parameters of the dispatchable generators.

Variables (columns):

- generator: number of generator (1-4)
- a1: linear cost parameter a_1
- a2: quadratic cost parameter a_2
- pmin: minimal generation p_{min} [MWh]
- pmax: maximal generation p_{max} [MWh]

DataC_hourlyload.csv

This file contains hourly load in Germany at the 7th of March 2023 (source: smard.de)

Variables (columns):

- Date: date of load (always 'Mar 7')
- Start: start of hour under consideration
- End: end of hour under consideration
- Total (grid load) [MWh] Calculated resolutions: grid load per hour [MWh]

DataC_nondispgeneration.csv

This file contains an artificial discrete probability distribution of the non-dispatchable generators.

Variables (columns):

- hour: block wise hour column, contains the values 0-5, 6-10, 11-14, 15-19, 20-23
- generation wind: four distinct possible wind generations for each hour block
- probability wind: probabilities corresponding to ‘generation wind’
- generation solar: four solar generations for each hour block
- probability solar: probabilities corresponding to ‘generation solar’